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Computational approaches for MS-based yeast metabolomics-centric multi-omics data integration: critical survey and perspectives

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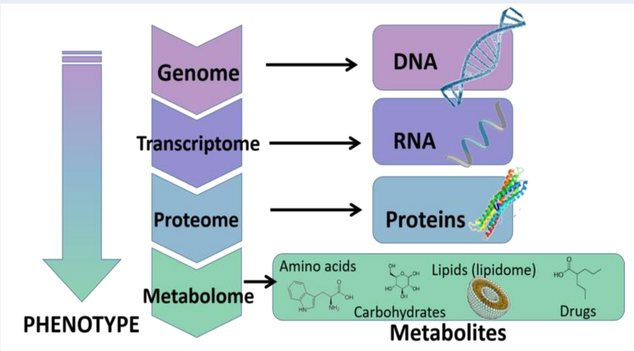
INTRODUCTION



Metabolism is a significant biological process, producing all biomolecules and energy for the cellular functions, while also interacting with the extracellular environment. The metabolites, which act as reactants and products of metabolic reactions, have also a regulatory function on both transcriptome and proteome.
Therefore, metabolomics together with genomics, transcriptomics and proteomics provides a systemic picture of the biological sample.

Metabolome characterization is accomplished by untargeted metabolomics studies. UHPLC coupled to Orbitrap Fusion™ Tribrid™ (Thermo Fisher Scientific) HRMS platform is one of the most efficient instruments to analyze and identify metabolites.

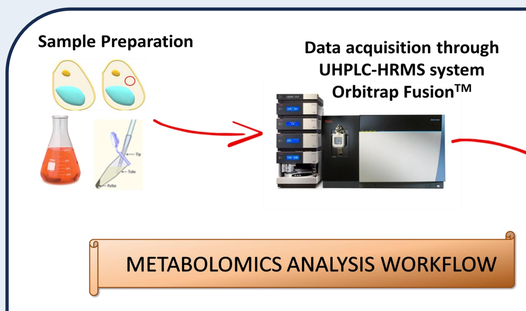
Yeast is an ideal model organism to study molecular and systems biology of higher eukaryotes, given the high similarity of cellular components and processes between yeast and humans. However, there is the need for standardization of yeast metabolome databases as well as development of bioinformatics tools for integrative omics analysis to gain a comprehensive understanding of the cell in a systems biology approach.

RESULTS

SURVEY OF OPEN-SOURCE TOOLS FOR METABOLOMICS DATA ANALYSIS AND MULTI-OMICS INTEGRATION			
REPOSITORY	MetaboLights	Metabolomics Workbench	MeRy-B
ANNOTATION	SIRIUS	MetaboAnalyst 5.0	
NETWORK ANALYSIS	NetID	CANOPUS	
REFENCE DATA DRIVEN ANALYSIS	MASST (GNPS)		
OMICS DATA INTEGRATION	MetaboAnalyst 5.0	Metaboverse	MOFA

Data previously produced by untargeted differential metabolome analysis of yeast ρ^0 e ρ^+ cells were used to test the performance of open-source software and to compare them with proprietary Compound Discoverer (CD)™ software (Thermo Fisher Scientific). CD™ is a Thermo-Fisher Scientific proprietary software that offers a fully integrated suite of advanced tools for metabolomics data processing and interpretation.

COMPOUND DISCOVERER™



METABOLOMICS ANALYSIS WORKFLOW

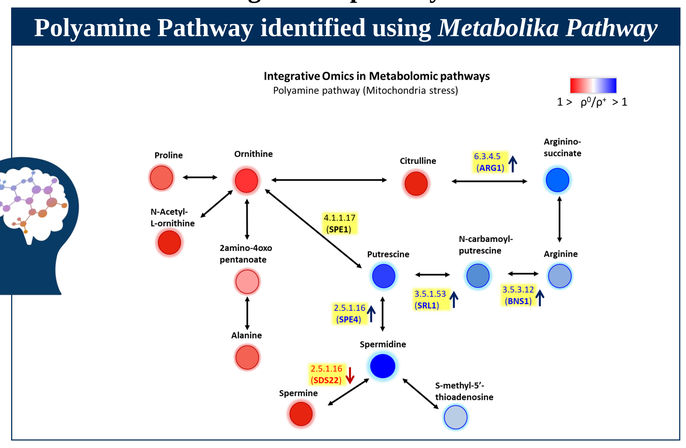
Differential metabolome	
Name	Log2 Fold Change: (Rho_0) / (Rho_piu)
Thiirene	0,6
ET8750000	-0,04
1-pyrroline	-0,5
2-Methylpropionitrile	-1,25
Pyrrolidine	2,2
3-Aminopropanal	-1,65

+

Differential transcriptome	
Name	Log2 Fold Change: (Rho_0) / (Rho_piu)
YAL001C	-1,072708025
YAL003W	-3,944214061
YAL004W	-0,561126332
YAL011W	-0,513920535
YAL013W	0,717376131
YAL016W	-1,644269201

Metabolic changes in respiratory deficient cells

Polyamine Pathway identified using *Metabolika Pathway*



We have set up an experimental system in which we compared metabolome in ρ^+ and ρ^0 prototrophic YSBN1 yeast cells. Untargeted metabolomics analysis was performed using the UHPLC-HRMS system Orbitrap Fusion™ Tribrid™ and data analysis software CD™. Through multivariate data analysis and comparison of the metabolomes, we were able to identify metabolites whose concentration changed due to mitochondrial dysfunction (ρ^0 yeast cells) (*see CD™ metabolite annotation below*).

Name	Formula	Annot. Source	Annot. ΔMass [ppm]	Calc. MW	m/z	RT [m	Area (Max.)
				198.81529	197.8080	4.513	8063240
				196.81834	195.8110	4.515	6314585
Cytidylyl-(3â€²â†’5â€²)-adenosine	C19 H25 N8 O11 P		0.75	572.13847	571.1311	4.640	1835264
	C2 H7 N O2 S2		0.52	140.99189	139.9846	4.689	11472849
PI(16:0/0:0)	C25 H49 O12 P		0.46	572.29643	571.2891	4.722	1344058
Inosine	C10 H12 N4 O5		-0.46	268.08065	267.0733	4.754	57052782
Hypoxanthine	C5 H4 N4 O		0.75	136.03861	135.0313	4.806	5593690
Dihydroxyacetone	C3 H6 O3		0.33	90.03172	89.02445	4.830	41879472
(2S)-2-Isopropyl-3-oxosuccinate	C7 H10 O5		-0.17	174.05279	173.0455	4.836	38265248
2,5-Didehydro-D-gluconate	C6 H8 O7		-0.12	192.02698	191.0197	4.845	178553365
Mesaconate	C5 H6 O4		1.20	130.02677	129.0194	4.855	23443756
Pentane-3,4-Diol-5-Phosphate	C5 H13 O6 P		1.07	200.04519	199.0379	4.901	5218800

The differential metabolomics data were integrated with differential transcriptomic profiles obtained with whole genome microarray analysis[1]. In a knowledge-based integration method, *Metabolika Pathway*, embedded within CD™, was used, exploiting the *Saccharomyces Genome Database (SGD)* to search for enzymes differentially expressed involved in the pathway. This allowed the identification of specific metabolic pathways altered in respiratory deficient ρ^0 cells- A typical altered pathway is shown (*see left panels*).

A comparison between this knowledge-based method for integration of metabolomics and transcriptomics data and a fully computer-aided integration method was made. Data produced by untargeted differential metabolome analysis of ρ^0 e ρ^+ cells, annotated and analyzed with CD™, were used to test the performance of open-source software.

Differential metabolome	
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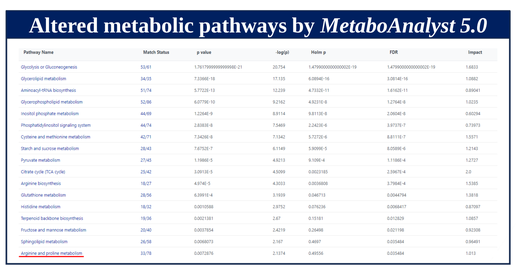
Altered metabolic pathways by *MetaboAnalyst 5.0*



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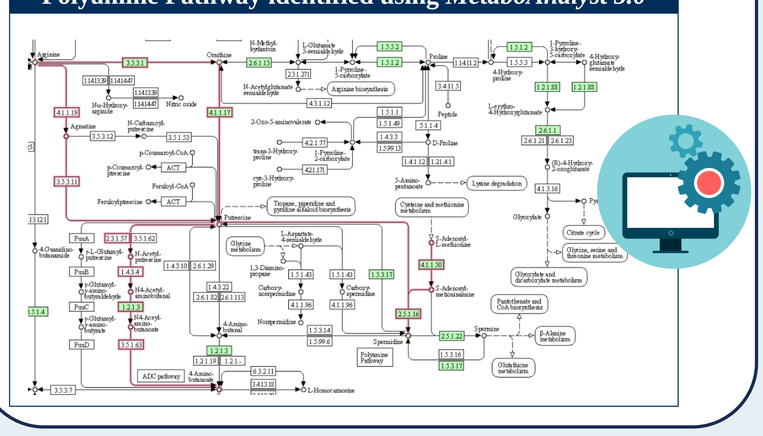
Polyamine Pathway identified using *MetaboAnalyst 5.0*



In order to use *Metaboanalyst 5.0* to integrate metabolomics with transcriptomics data, it was necessary to check compound and gene nomenclature to make the software run correctly. The results obtained were virtually comparable to those obtained through knowledge-based data integration (*see right panels*). *MetaboAnalyst 5.0* shows only superpathways, which include a connected set of smaller pathways that share common metabolites. In this case, the right panel shows the *arginine and proline metabolism superpathway*, containing the *polyamine pathway* highlighted in red.

1, Epstein et al. Mol Biol Cell. 12 (2001) 297–308

CONCLUSIONS

- **The open-source tools currently available for the analysis of metabolomics data do not easily communicate with each other, it is therefore necessary to develop a pipeline to automate their use.**
- **Data produced by proprietary software, such as Compound Discoverer™, cannot be easily queried by open-source tools without a data modification process.**
- **Currently available open-source software is primarily based on human databases; therefore, it is necessary to develop new R packages, following FAIR principles, for metabolite annotation and integration of omics data with the possibility of selecting curated databases of yeasts and other organisms.**

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